# MemberPass

Smart Studio Membership & Access Verification System

## PROBLEM STATEMENT

Studio owners often struggle to manage memberships manually and prevent expired or inactive members from accessing studio facilities. Paper-based or honour-system approaches lead to revenue loss, security gaps and administrative overhead.

## HOW MIGHT WE?

*"How might we help studio owners ensure that only members with an active membership plan can securely enter their studio through a simple and reliable digital verification system?"*

## TARGET USERS

- Studio owners and administrators
- Receptionists and verification staff
- Studio members

## FIVE USER ACTIONS

1. Register a new studio member.
2. Assign an active membership plan.
3. Generate a digital membership card with QR code.
4. Scan the QR code at the studio entrance.
5. Automatically allow or deny entry based on membership validity.

## PROPOSED SOLUTION

Develop a responsive web application that manages studio memberships and uses secure QR-based verification to control studio entry automatically and reliably.

## TECHNOLOGY STACK

- React.js + Vite
- Tailwind CSS v4
- Node.js + Express.js
- MongoDB + Mongoose
- JWT Auth + bcryptjs
- qrcode / html5-qrcode

## EXPECTED OUTCOME

A functional digital membership system that reduces manual verification, prevents unauthorized entry and simplifies studio membership management for gyms and fitness studios of all sizes.